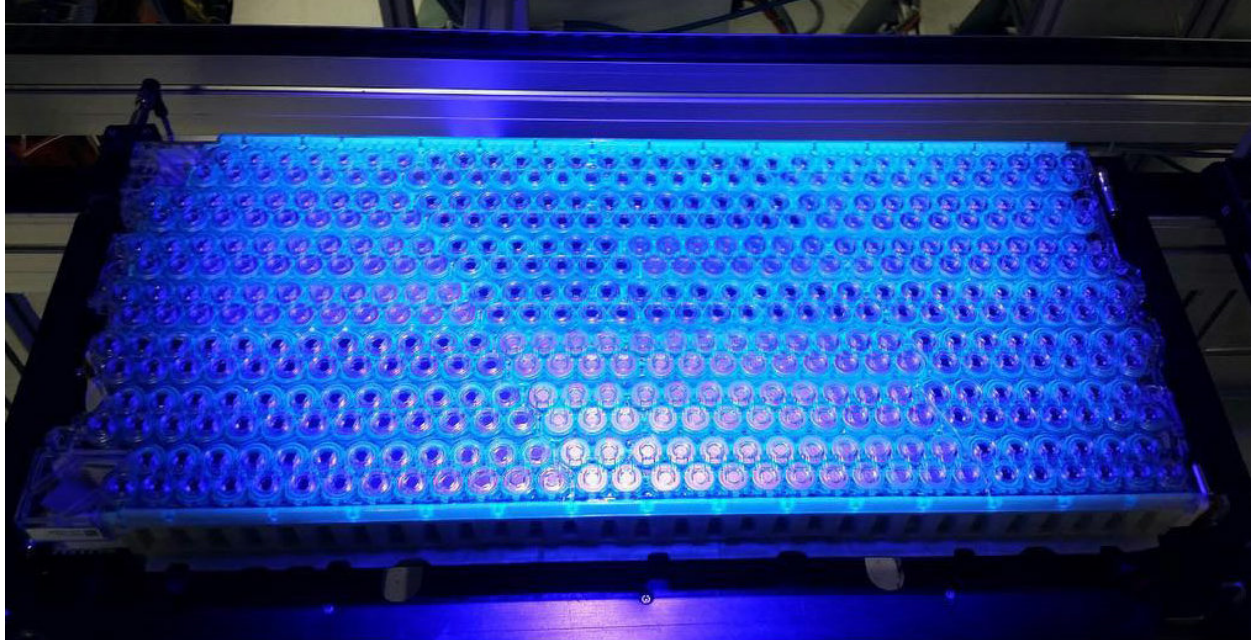


Battery capacity

Exam exercise: Battery capacity



In this exercise you will study a data set on battery capacity collected by researchers from AAU Energy. The batteries were kept at three different temperatures. The capacity loss Q (in percentage of initial capacity) is the response variable. We are interested in the effect on the response of the temperature and the efficient number of charge cycles FEC that the battery had been exposed to.

In the analysis, we will work with the log transformed variables $\log Q$ and $\log \text{FEC}$. Moreover, we will consider **Temperature** as a categorical variable with the three levels 35° , 40° , and 45° Celcius.

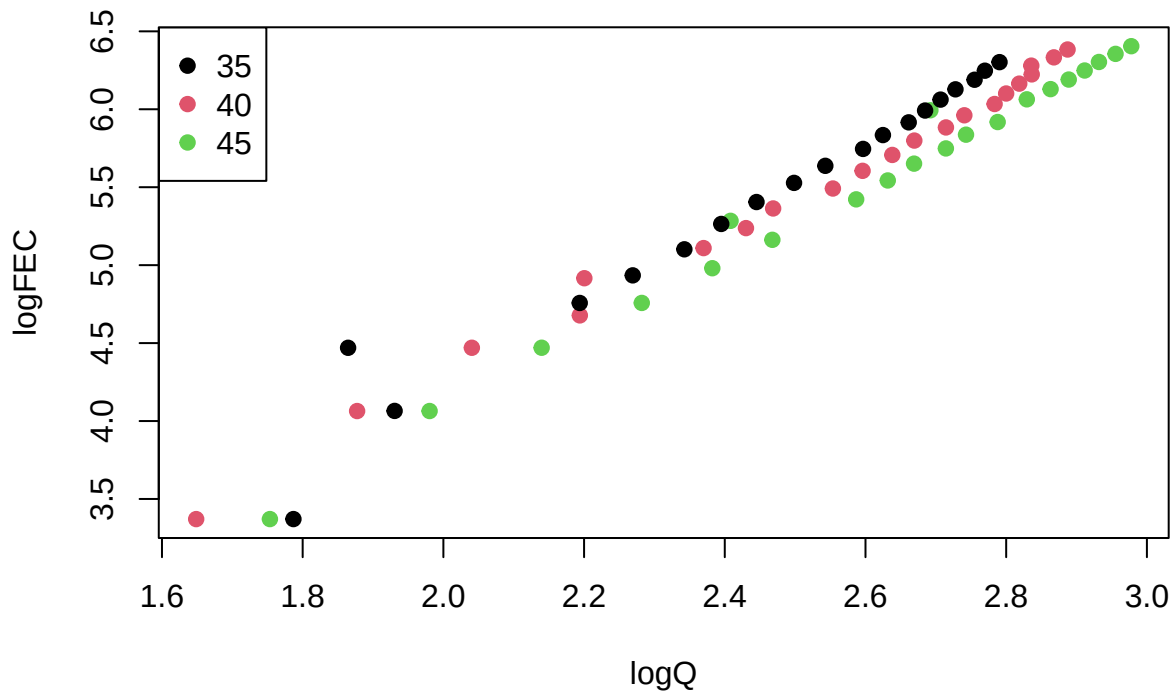
We begin by loading the **mosaic** package and reading in the data. Since we consider **Temperature** a categorical variable, we make this variable a factor.

```
library(mosaic)
capacity<-read.delim("https://asta.math.aau.dk/datasets?file=capacity.txt",sep="")
capacity$Temperature<-factor(capacity$Temperature)
```

1. Make a scatter plot of the log transformed variables $\log Q$ and $\log \text{FEC}$ where points are colored by temperature level.

```
plot(capacity$logQ, capacity$logFEC,
     col = capacity$Temperature, #categories
     pch = 19, #symbol for markers
     xlab = "logQ",
     ylab = "logFEC")
legend("topleft", legend = levels(capacity$Temperature),
```

```
col = 1:length(levels(capacity$Temperature)), pch = 19)
```



2. The code below finds the correlation between $\log Q$ and $\log \text{FEC}$ for each level of temperature. Explain the concept of correlation and interpret the results in relation to the plot from Question 1.

```
tempGrupper=split(capacity, capacity$Temperature)
lapply(tempGrupper, function(x) cor(x$logQ, x$logFEC))
```

```
## $`35`
## [1] 0.9813515
##
## $`40`
## [1] 0.9967649
##
## $`45`
## [1] 0.9943055
```

The absolute value of R (correlation) measures the (linear) strength of dependence between $\log Q$ and $\log \text{FEC}$. Since all three values are very close to 1, we can assume that $\log Q$ and $\log \text{FEC}$ are very correlated.

3. Consider a multiple regression model without interaction with $\log Q$ as the response variable and $\log \text{FEC}$ and Temperature as predictors. Write out the model equation using dummy variables. What is the interpretation of the parameters?

$$\log(Q) = \alpha + \beta_1 \cdot \log(\text{FEC}) + \beta_2 \cdot T + \epsilon$$

- α is the intercept. It corresponds to the value of $\log(Q)$ when both $\log(\text{FEC})$ and T are 0.
- β_1 and β_2 describe the change in the mean response ($\log(Q)$), when $\log(\text{FEC})$ or T are changed one unit respectively.
- ϵ is an error with a $\text{norm}(0, \sigma)$ distribution.

4. Edit the code below to fit the model from Question 3.

```
#tempArray <- as.numeric(as.character(capacity$Temperature))
model<-lm(logQ ~ logFEC + factor(capacity$Temperature),data=capacity)
summary(model)

##
## Call:
## lm(formula = logQ ~ logFEC + factor(capacity$Temperature), data = capacity)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.197468 -0.004855  0.009251  0.017714  0.176427
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.224846   0.042060    5.346 1.65e-06 ***
## logFEC           0.411014   0.007511   54.720 < 2e-16 ***
## factor(capacity$Temperature)40 0.043310   0.014654    2.956 0.00453 **
## factor(capacity$Temperature)45 0.106577   0.014665    7.268 1.14e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04625 on 57 degrees of freedom
## Multiple R-squared:  0.9819, Adjusted R-squared:  0.981
## F-statistic: 1032 on 3 and 57 DF,  p-value: < 2.2e-16
```

Explain the output from the code. Your explanation should as a minimum include:

- What is the prediction equation?

$$\log(Q) = \alpha + \beta_1 \cdot \log(\text{FEC}) + \beta_2 \cdot T + \epsilon$$

$$\log(Q) = -0.152700 + 0.410957 \cdot \log(\text{FEC}) + 0.010693 \cdot T + \epsilon$$

- Explain the calculation of t value and determination and interpretation of the p-value.

t value is calculated as $t = \frac{\text{Estimate}}{\text{Std. Error}}$.

The p-value is derived from the t-value using the t-distribution, which depends on the degrees of freedom. It quantifies the probability of getting a t-value that big (or bigger) if the true effect were actually zero.

- What is the interpretation of Multiple R-squared?

It measures the proportion of the variation in the response variable (logQ) that is explained by the predictors (logFEC and Temperature) in the model.

- Make an overall F-test for the null-hypothesis that there is no effect of any of the predictors.

```
anova(model)

## Analysis of Variance Table
##
## Response: logQ
##              Df Sum Sq Mean Sq  F value    Pr(>F)
## logFEC         1  6.5099   6.5099 3042.740 < 2.2e-16 ***
## factor(capacity$Temperature) 2  0.1152   0.0576   26.926 5.851e-09 ***
## Residuals     57  0.1220   0.0021
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F value is the ratio of explained variance to unexplained variance and $\Pr(>F)$ is the p-value for the F-test of each predictor. From the values, we can reject the null hypothesis. At least one predictor (actually both in this case) significantly affects $\log(Q)$.

- Investigate whether or not there is an interaction between the effect of Temperature and the effect of logFEC as predictors of logQ (Hint: With more than two groups you need to use the `anova` function to make the test, see p. 31 in the slides from the second lecture of the module).

```
model_interaction <- lm(logQ ~ logFEC * factor(capacity$Temperature), data=capacity)
summary(model_interaction)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC * factor(capacity$Temperature), data = capacity)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.222724	-0.001361	0.009642	0.015118	0.121917

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.36911	0.07107	5.193	3.1e-06 ***
logFEC	0.38439	0.01298	29.609	< 2e-16 ***
factor(capacity\$Temperature)40	-0.21810	0.09817	-2.222	0.03045 *
factor(capacity\$Temperature)45	-0.04996	0.09829	-0.508	0.61326
logFEC:factor(capacity\$Temperature)40	0.04798	0.01783	2.691	0.00942 **
logFEC:factor(capacity\$Temperature)45	0.02885	0.01780	1.621	0.11080

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04425 on 55 degrees of freedom
## Multiple R-squared:  0.984, Adjusted R-squared:  0.9826
## F-statistic: 678.3 on 5 and 55 DF,  p-value: < 2.2e-16
```

```
anova(model, model_interaction)
```

```
## Analysis of Variance Table
##
## Model 1: logQ ~ logFEC + factor(capacity$Temperature)
## Model 2: logQ ~ logFEC * factor(capacity$Temperature)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      57 0.12195
## 2      55 0.10767  2  0.014278 3.6465 0.03257 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comparing the model without interaction to the model with interaction:

F = 2.3183, p = 0.1334, not significant.

Adding the interaction does not significantly improve the model. There is no significant interaction between Temperature and logFEC.

- Plot the model with and without interaction and explain the difference.

```
library(ggplot2)
```

```
# make sure Temperature is a factor
```

```
capacity$Temperature <- factor(capacity$Temperature)

# Fit models using the factor variable
model_no_interaction <- lm(logQ ~ logFEC + Temperature, data = capacity)
model_interaction <- lm(logQ ~ logFEC * Temperature, data = capacity)

summary(model_no_interaction)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC + Temperature, data = capacity)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.197468	-0.004855	0.009251	0.017714	0.176427

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.224846	0.042060	5.346	1.65e-06 ***
logFEC	0.411014	0.007511	54.720	< 2e-16 ***
Temperature40	0.043310	0.014654	2.956	0.00453 **
Temperature45	0.106577	0.014665	7.268	1.14e-09 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04625 on 57 degrees of freedom
## Multiple R-squared:  0.9819, Adjusted R-squared:  0.981
## F-statistic: 1032 on 3 and 57 DF,  p-value: < 2.2e-16
```

```
summary(model_interaction)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC * Temperature, data = capacity)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.222724	-0.001361	0.009642	0.015118	0.121917

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.36911	0.07107	5.193	3.1e-06 ***
logFEC	0.38439	0.01298	29.609	< 2e-16 ***
Temperature40	-0.21810	0.09817	-2.222	0.03045 *
Temperature45	-0.04996	0.09829	-0.508	0.61326
logFEC:Temperature40	0.04798	0.01783	2.691	0.00942 **
logFEC:Temperature45	0.02885	0.01780	1.621	0.11080

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04425 on 55 degrees of freedom
## Multiple R-squared:  0.984, Adjusted R-squared:  0.9826
## F-statistic: 678.3 on 5 and 55 DF,  p-value: < 2.2e-16
```

```

# Compare models (test for interaction)
anova(model_no_interaction, model_interaction)

## Analysis of Variance Table
##
## Model 1: logQ ~ logFEC + Temperature
## Model 2: logQ ~ logFEC * Temperature
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      57 0.12195
## 2      55 0.10767  2  0.014278 3.6465 0.03257 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Create grid for predictions using Temperature levels (same names as model)
newdat <- expand.grid(
  logFEC = seq(min(capacity$logFEC, na.rm = TRUE),
               max(capacity$logFEC, na.rm = TRUE),
               length.out = 200),
  Temperature = levels(capacity$Temperature)
)

# Predict from both models (now newdat matches the model variables)
newdat$pred_no_interaction <- predict(model_no_interaction, newdata = newdat)
newdat$pred_interaction <- predict(model_interaction, newdata = newdat)

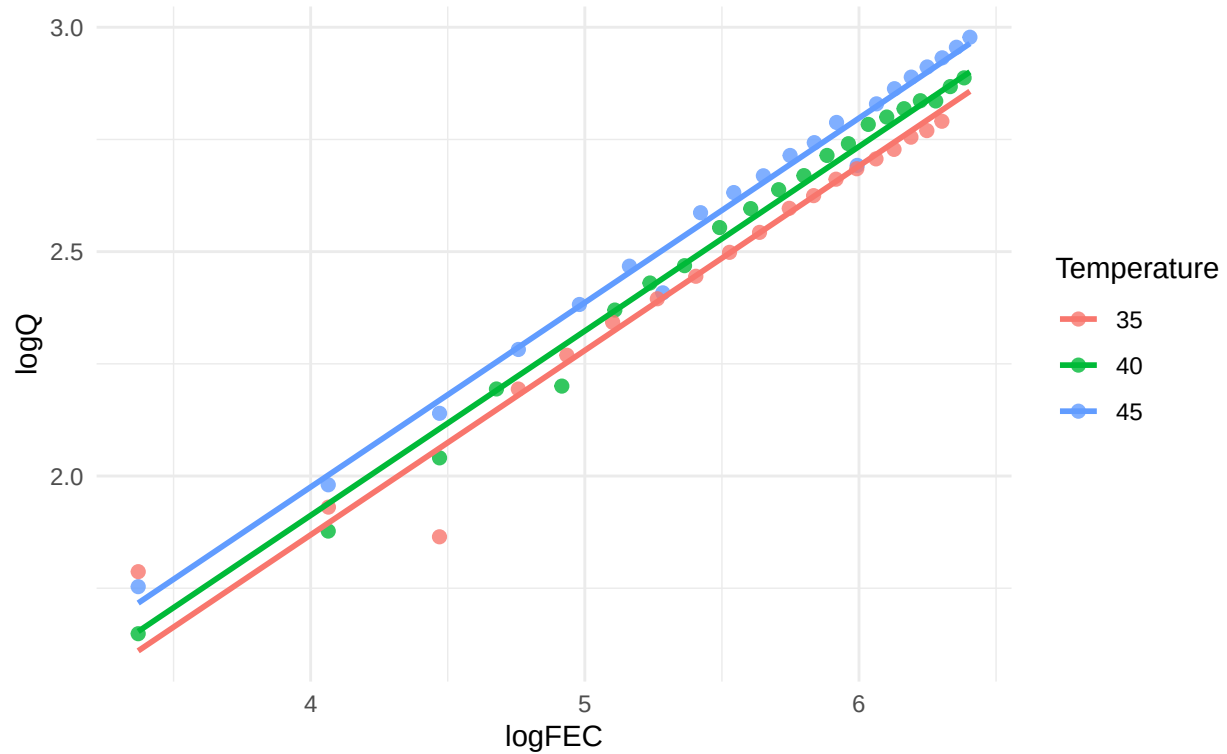
ggplot(capacity, aes(x = logFEC, y = logQ, color = Temperature)) +
  geom_point(alpha = 0.8, size = 2) +
  geom_line(
    data = newdat,
    aes(y = pred_no_interaction),
    size = 1
  ) +
  labs(
    title = "No-interaction model",
    subtitle = "Common slope across temperatures (parallel lines)",
    x = "logFEC",
    y = "logQ"
  ) +
  theme_minimal()

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

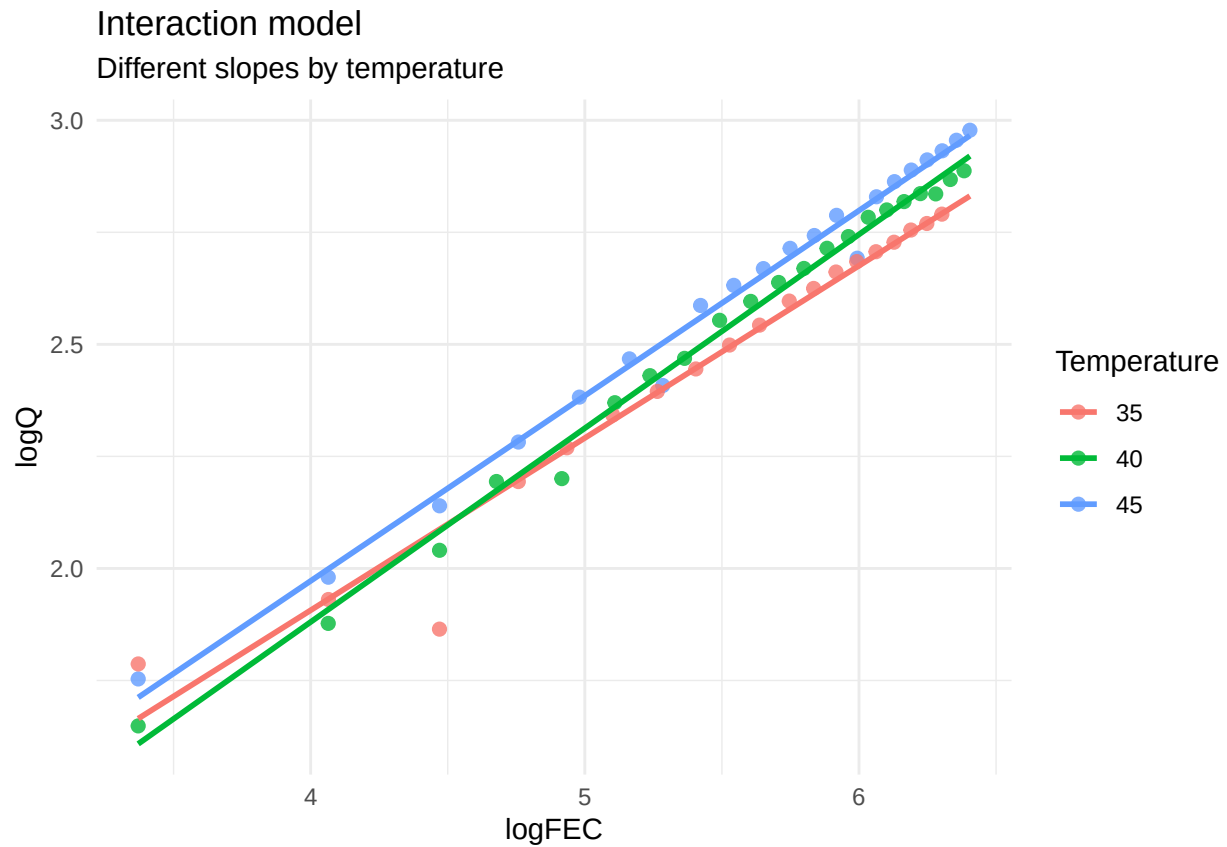
```

No-interaction model

Common slope across temperatures (parallel lines)

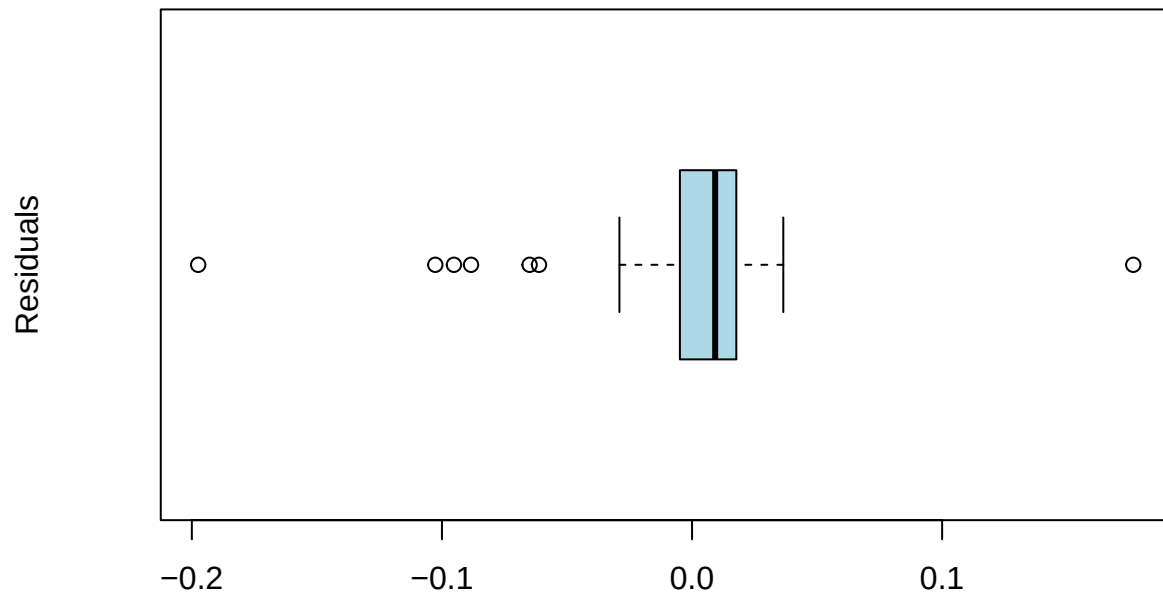


```
ggplot(capacity, aes(x = logFEC, y = logQ, color = Temperature)) +  
  geom_point(alpha = 0.8, size = 2) +  
  geom_line(  
    data = newdat,  
    aes(y = pred_interaction),  
    size = 1  
  ) +  
  labs(  
    title = "Interaction model",  
    subtitle = "Different slopes by temperature",  
    x = "logFEC",  
    y = "logQ"  
  ) +  
  theme_minimal()
```



```
# Residual boxplot for the no-interaction model
residuals_no_int <- residuals(model_no_interaction)
boxplot(residuals_no_int,
        main = "Boxplot of residuals (no-interaction model)",
        ylab = "Residuals",
        col = "lightblue", horizontal = TRUE)
abline(h = 0, col = "red", lty = 2)
```


Boxplot of residuals (no-interaction model)



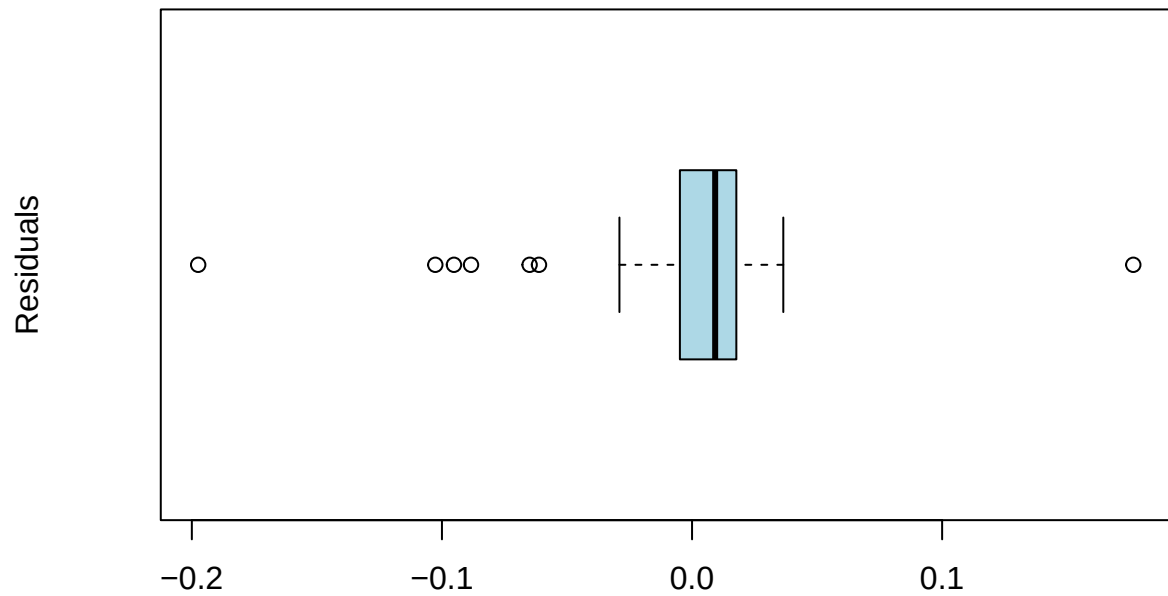
7. The code below computes the residuals $y_i - \hat{y}_i$ in the model without interaction. Draw a boxplot of the residuals and comment on the result.

```
# compute residuals
residuals <- model$residuals # or m0$residuals if using the variable from previous chunk

# create a boxplot of residuals
boxplot(residuals,
  main = "Boxplot of residuals (no-interaction model)",
  ylab = "Residuals",
  col = "lightblue",
  horizontal = TRUE)

# optionally, add a horizontal line at 0
abline(h = 0, col = "red", lty = 2)
```

Boxplot of residuals (no-interaction model)



The median residual (thick line in the box) is very close to 0, which suggests that the model predictions are roughly unbiased on average. There are several points outside the whiskers, on both the left and right sides. Negative outliers (~ -0.2 to -0.1) indicate that the model overestimates $\log Q$ for these observations.